FAQs- SFM-01

1. Can we calculate the exponential moving average on Excel?

Yes. This will be covered in one of the upcoming lectures' assignments.

2. I don't have MS Excel. Can I use Google Sheets online?

You can use Microsoft Excel, LibreOffice Calc, or OpenOffice Calc for most of the calculations we do in various lectures. However, there would be some syntactical and operational differences. The instructor would be using Microsoft Excel.

3. What are the sources for tick-by-tick (TBT) data?

Tick-by-tick data is generally used when High-Frequency Trading (HFT) is the objective. Such data can be subscribed to directly from the exchanges or an exchange-authorized data vendor.

4. Why is Adj Close different from the Close price every day in the data we use in class? Was there a corporate action every day?

Adjusted close price reflects the effect of various corporate actions (dividend payment, splits, and so on) carried out on the stocks in the past. Why do we need to adjust the close price? If we don't adjust the data to reflect the historical corporate actions, the data would have gaps in the prices, and it would lead to incorrect backtesting results. No, there is not a corporate action taking place each day; however, the effect of one corporate action would impact all future prices, and hence, the need for adjusting all subsequent prices.

5. Why don't we have Adjusted Low/High?

We can also have adjusted open, high, and low prices. We can calculate them if we have corporate action data or at least one adjusted price variable.

6. When to use the dollar sign ($) in the Excel formula?

The dollar sign ($) is used to fix the reference to the row and/or column in a formula. Suppose we have a formula in some cell that is "= A1". To fix the column reference, we would prefix the column name 'A' with the '$' sign. The formula would thus be modified to "=$A1". Similarly, to fix the row name, we would type "=A$1". If we want to fix both, we can type '=A1'.

7. If Excel can calculate most of the backtesting operations, then what are the benefits of using Python?

Excel is a good platform to start with. It works quite well when we have relatively small datasets. However, it is also time-consuming to backtest the same strategy on multiple stocks in Excel. Python overcomes such issues, along with many more. Please refer to this webinar for a detailed comparison: https://blog.quantinsti.com/stock-data-analysis-excel-python-23-february-2021/

8. If we are backtesting a strategy that includes OHLCV data, should we apply the adjustment factor and adjust the data, or not? How to decide?

It depends on the source of the data. If the data you sourced is already adjusted, you can use it directly. For example, Yahoo Finance provides split and dividend-adjusted data in the 'Adj Close' column. Whereas, if we source data directly from the exchange, we usually get unadjusted data. Please verify the data with the vendor before making any adjustments.

9. We can get VWAP, ATR, and other technical indicators directly from Reuters/Bloomberg and other platforms, so why do we do them manually here?

One of the objectives of the class is to understand how we can use tools like Excel to validate a trading idea. Additionally, this class enables us to calculate those indicators systematically and explains the rationale and calculations behind them.

10. From which date should we backtest our strategy?

The purpose of backtesting a strategy is to validate our trading idea. To confidently validate it, we should use as much relevant data as possible.

11. What corporate actions would impact the stock price and need to be adjusted in the close price?

Generally, stock prices are adjusted for stock splits and dividends. However, other corporate actions like bonus issues, rights issues, mergers and acquisitions, and spin-offs can also impact the price and may require adjustments for accurate historical analysis.

12. I do not have a licensed version of Excel - What is the way out for me?

You can use LibreOffice Calc, OpenOffice Calc, or Google Sheets for most of the calculations we do in various lectures. However, there would be some syntactical and operational differences as compared to MS Excel that the instructor would be using.

13. Is it possible to build a strategy that always works? Has anybody done that?

Nothing works permanently. Everything changes, and that's the way the market works as well. Even strategies have a shelf life. Hence, in a professional setting, multiple strategies are employed (diversification in strategy), existing strategies are regularly validated, and a pipeline of strategies is developed for the future.

14. Are adjusted close available if the data is downloaded intraday, when markets are still active?

Adjusted data is usually available on a daily basis. Corporate actions-related adjustments don’t get incorporated during live markets.

15. Does the ATR approach covered in the class work for other strategies, as there are many trading strategies that one can model, which might be very different from the one we saw in the class?

We learned how we can model a complex quantitative strategy using Excel. We broke down each component of the strategy: data sanity check, building indicators, calculating entry (trigger price), and exit (SL and TP for both long and short positions) rules. Towards the end of the lecture, we'd seen how we can optimize a strategy. One can use a similar approach to build a strategy by breaking down each component and listing all the rules of a strategy.

16. What is the Average True Range (ATR)?

ATR is a technical indicator that captures the range of price levels of an asset over a specified period. As the name suggests, it gives us an average value of the price range. It measures the volatility of an asset, taking into account any gaps in the price movement.

17. How would the modelling steps change if we change the data frequency in the ATR-based breakout strategy, such as to 5 mins or 10 mins?

The approach would remain pretty much the same. We need to ensure that we are not inadvertently violating the rules laid out for a strategy. For example, we need to make sure that we are not introducing any look-ahead bias. Or, if the strategy rules dictate that the positions should be squared off at market close, we must adhere to them.

18. In the lecture, we performed all analyses on historical data. How do we implement it in live trading?

The process goes like this:

- You build a trading hypothesis.
- Backtest it on the historical data.
- Optimize the parameters.
- Develop the execution algorithm
- Deploy it in a live market

In the lecture, we completed the first three points. In the classes to come, you will learn the rest of the steps.

19. What process do we follow to optimize the parameters to maximize the profitability of a strategy?

If we want to optimize two parameters, we can use a data table to do so. For more parameters, it would be somewhat challenging to optimize in Excel. We can use a programming language like Python to optimize two or more parameters to maximize a strategy's profitability.

20. By when would the in-class files be uploaded on the LMS?

In-class files will be uploaded on the LMS under the SFM module within 24 hours after the lecture ends.